

(A0726772)

Patient MRN: 007087315 | Gender: Male

Diagnosis: Lung adenocarcinoma | Test Number 1

Therapy Finder Page

REPORTING

Report Date: APR-19-2023
Receipt Date: APR-14-2023
Collection Date: APR-13-2023
Specimen: Blood
Status: FINAL

PHYSICIAN

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Additional Recipient: N/A



Complete Tumor Response Map on page 2

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Summary of Detected Somatic Alterations, Immunotherapy Biomarkers & Associated Treatment Options

KEY Approved in indication Approved in other indication Lack of response

Detected Alteration(s) / Biomarker(s)	Associated FDA-approved therapies	Clinical trial availability (see page 3)	% cfDNA or Amplification
ALK L1196Q	Alectinib, Crizotinib	Yes	0.4%
EML4-ALK Fusion	Brigatinib, Ceritinib, Lorlatinib	Yes	0.2%
CDKN2A R80*	None	Yes	0.2%

Variants of Uncertain Clinical Significance

ALK-EML4 Fusion (0.5%)

The functional consequences and/or clinical significance of alterations are unknown. Relevance of therapies targeting these alterations is uncertain.

Comments

In this patient, both a EML4-ALK fusion and a ALK-EML4 fusion were detected in circulating cell-free DNA. The ALK-EML4 fusion represents the reciprocal gene rearrangement of the EML4-ALK gene rearrangement and is detected in circulating cell-free DNA in this patient. The clinically significant rearrangement for this patient is the EML4-ALK fusion.

Additional Biomarkers

Biomarker	Additional Details
MSI-High	NOT DETECTED

Alterations or biomarkers that were "**NOT DETECTED**" have been excluded from the summary table above.

We evaluated this sample for 74 genes, including the following guideline-recommended genes for NSCLC

EGFR(T790M and others)

ALK

ROS1

BRAF

MET

ERBB2(HER2)

RET

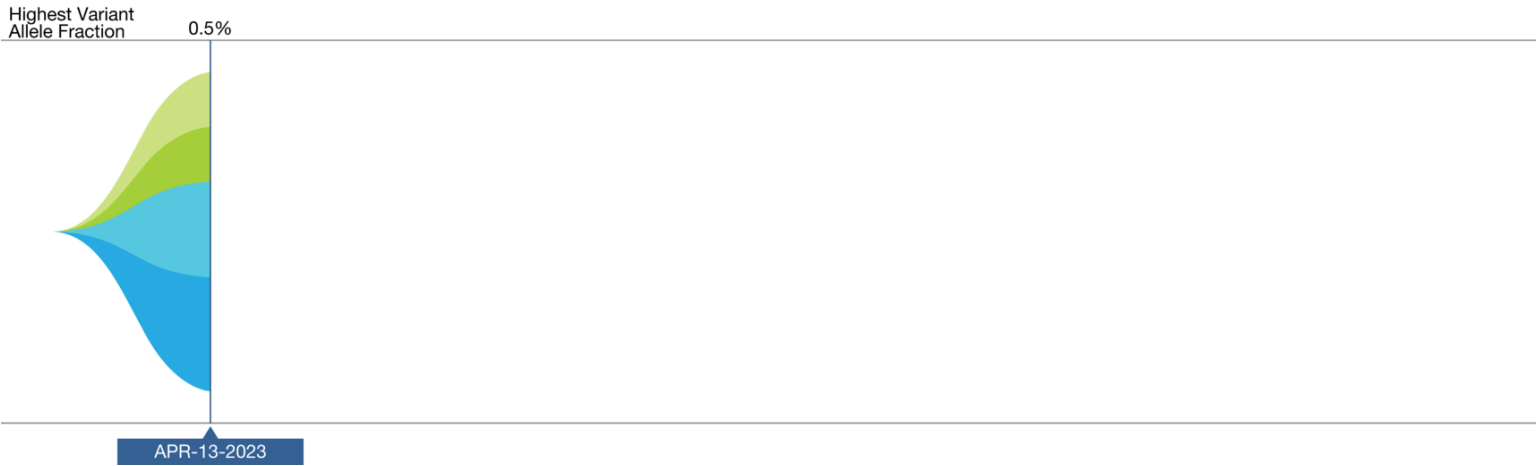
NTRK

KRAS

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Guardant360 Tumor Response Map

The Guardant360 laboratory developed test (LDT) Tumor Response Map illustrates the variant allele fraction (% cfDNA) of observed somatic variants at each sample submission. Amplifications are not plotted, and only the first and last five test dates are plotted. Please see the Physician Portal (portal.guardanthealth.com) for the Tumor Response Map with all test dates.



Detected Alteration(s) / Biomarker(s)	% cfDNA or Amp	
ALK- <i>EML4</i> Fusion	0.5%	Variants of Uncertain Clinical Significance §
ALK L1196Q	0.4%	
<i>EML4</i> -ALK Fusion	0.2%	
<i>CDKN2A</i> R80*	0.2%	

The table above annotates the variant allele fraction (% cfDNA) detected in this sample, listed in descending order.
§ See definitions section for more detail

Available Clinical Trials (within the same state as the ordering physician)

There may be additional trials not listed here. Visit: portal.guardanthealth.com or email clientservices@guardanthealth.com with A0726772 in the subject line of the email, for additional trials.

Alteration	Trial ID / Contact	Title	Phase	Site(s)
CDKN2A R80*	NCT04553133 Pfizer CT.gov Call Center,ClinicalTrials.gov_Inquiries@pfizer.com,1-800-718-1021	PF-07104091 as a Single Agent and in Combination Therapy	Phase 1 /Phase 2	Louisville, KY (11)
	Visit portal.guardanthealth.com for trials not within the same state as the physician's office			
ALK L1196Q	Visit portal.guardanthealth.com for trials not within the same state as the physician's office			
EML4-ALK Fusion	Visit portal.guardanthealth.com for trials not within the same state as the physician's office			

More clinical trial options available at portal.guardanthealth.com

Definitions

Variants of Uncertain Clinical Significance: The functional consequences and/or clinical significance of alterations are unknown. Relevance of therapies targeting these alterations is uncertain.

Fusion: Fusion events are gene rearrangements that fuse two previously distinct genes into a single genetic unit. Guardant360 detects fusions in the genes listed in Table 1.

***Nonsense mutation:** A point mutation that results in a premature stop codon.

Interpretation

Somatic alterations were detected in the circulating cell-free DNA isolated from this patient's blood specimen. These genomic alterations are cancer-associated somatic variants, some of which have been associated with either increased or reduced clinical response to specific treatments. The percentage of altered cell-free DNA circulating (% cfDNA) in blood is related to the unique tumor biology of each patient. Factors that may affect the % cfDNA of detected somatic alterations include tumor growth, turn over, size, heterogeneity, vascularization, disease progression, and treatment.

Method and Limitations

Guardant360 sequences 74 cancer-associated genes to identify somatic alterations. Cell-free DNA (cfDNA) is extracted from plasma, enriched for targeted regions, and sequenced using the Illumina platform and hg19 as the reference genome. All exons are sequenced in some genes; only clinically significant exons are sequenced in other genes. The types of genomic alterations detected by Guardant360 include single nucleotide variants, gene amplifications, fusions, short insertions/deletions (longest detected, 70 base pairs), and splice site disrupting events (see Table 1). Microsatellite Instability (MSI) is assessed for all cancer types by evaluating somatic changes in the length of repetitive sequences on the Guardant360 panel. A "Not Detected" result in samples where the highest % cfDNA is < 0.2% is an inconclusive result because it does not preclude MSI-High status in tissue. MSI status is currently not reported for earlier panel versions. This version of the Guardant360 test is not validated for the detection of other types of genomic alterations, such as complex rearrangements or gene deletions. Certain sample or variant characteristics, such as low cfDNA concentration, may result in reduced analytic sensitivity. Guardant360 cannot discern the source of circulating cfDNA, and for some variants in the range of ~40 to 60% cfDNA, the test cannot easily distinguish germline variants from somatic alterations. Guardant360 is not validated for the detection of germline or de novo variants that are associated with hereditary cancer risk. Tissue genotyping should be considered when plasma genotyping is negative, if clinically appropriate.

Table 1: Genes on the Guardant360 Panel

Guardant360 reports single nucleotide variants, splice site mutations, and insertion and deletion variants (indels) in all clinically relevant exons in 74 genes and reports other variant types in select genes as indicated below.

<i>AKT1</i>	<i>ALK</i> #	<i>APC</i>	<i>AR</i> †	<i>ARAF</i>	<i>ARID1A</i>	<i>ATM</i>	<i>BRAF</i> †	<i>BRCA1</i>
<i>BRCA2</i>	<i>CCND1</i> †	<i>CCND2</i> †	<i>CCNE1</i> †	<i>CDH1</i>	<i>CDK12</i>	<i>CDK4</i> †	<i>CDK6</i> †	<i>CDKN2A</i>
<i>CTNNB1</i>	<i>DDR2</i>	<i>EGFR</i> †	<i>ERBB2</i> †	<i>ESR1</i>	<i>EZH2</i>	<i>FBXW7</i>	<i>FGFR1</i> †	<i>FGFR2</i> # †
<i>FGFR3</i> #	<i>GATA3</i>	<i>GNA11</i>	<i>GNAQ</i>	<i>GNAS</i>	<i>HNF1A</i>	<i>HRAS</i>	<i>IDH1</i>	<i>IDH2</i>
<i>JAK2</i>	<i>JAK3</i>	<i>KIT</i> †	<i>KRAS</i> †	<i>MAP2K1</i>	<i>MAP2K2</i>	<i>MAPK1</i>	<i>MAPK3</i>	<i>MET</i> †
<i>MLH1</i>	<i>MPL</i>	<i>MTOR</i>	<i>MYC</i> †	<i>NF1</i>	<i>NFE2L2</i>	<i>NOTCH1</i>	<i>NPM1</i>	<i>NRAS</i>
<i>NTRK1</i> #	<i>NTRK3</i>	<i>PDGFRA</i> †	<i>PIK3CA</i> †	<i>PTEN</i>	<i>PTPN11</i>	<i>RAF1</i> †	<i>RB1</i>	<i>RET</i> #
<i>RHEB</i>	<i>RHOA</i>	<i>RIT1</i>	<i>ROS1</i> #	<i>SMAD4</i>	<i>SMO</i>	<i>STK11</i>	<i>TERT</i> ‡	<i>TP53</i>
<i>TSC1</i>	<i>VHL</i>							

‡ Guardant360 reports alterations in the promoter region of this gene.

Guardant360 reports fusion events involving this gene for all known gene partners.

† Guardant360 reports amplifications of this gene.

About the Test

The Guardant360 assay was developed and its performance characteristics were determined by Guardant Health, Inc. This test has not been cleared or approved by the U.S. Food and Drug Administration (FDA). The FDA has determined that such clearance or approval is not necessary. This test may be used for clinical purposes and should not be regarded as investigational or for research only. Guardant Health's clinical reference laboratory is certified under the Clinical Laboratory Improvement Amendments of 1988 (CLIA) as qualified to perform high complexity clinical laboratory testing. The laboratory report should be interpreted in the context of other clinical information and laboratory, pathology, and imaging studies by a qualified medical professional prior to initiating or changing a patient's treatment plan. The selection of any, all, or none of the drugs associated with potential clinical benefit (or potential lack of clinical benefit) is entirely at the discretion of the treating medical professional. Drug and trial information are based on the diagnosis written on the submitted test request form; this information is not based on any supplemental information provided by the requesting medical professional, including pathology reports or other molecular studies. Some drugs listed in this report may not be approved or cleared by the FDA for the indicated use. Guardant Health makes no endorsement, express or implied, of any product, physician, or procedure contained in this report. This report makes no promises or guarantees that a particular medication will affect (or not affect) the clinical outcome of any patient.

Testing performed at: Guardant Health

Laboratory Director: Martina Lefterova, MD PhD | CLIA ID: 05D2070300 | CAP #: 8765297 | 505 Penobscot Drive, Redwood City, CA, 94063, USA

Additional information is available

Any therapeutic annotations are based on publicly available information. This information is described in the "Detailed Therapy Results" and "Relevance of Detected Alterations" sections.

Visit portal.guardanthealth.com or email clientservices@guardanthealth.com with A0726772 in the subject line of the email for:

- Additional clinical trials
- Detailed Therapy Results
- Relevance of Detected Alterations
- References

If you would like to receive this additional information with every Guardant360 report, please call client services at [855.698.8887](tel:855.698.8887) to opt-in.

Additional information begins on the next page.

List of Available Clinical Trials

Alteration	Trial ID / Contact	Title	Phase	Site (number in parenthesis is count of trial sites)
CDKN2A R80*	NCT03449381 Boehringer Ingelheim,clintriage.rdg@boehringer-ingelheim.com,1-800-243-0127	This Study Aims to Find the Best Dose of BI 907828 in Patients With Different Types of Advanced Cancer (Solid Tumors)	Phase 1	Houston, TX; Detroit, MI; Madison, WI; New York, NY; New Haven, CT; Sarasota, FL; Nashville, TN; Santa Monica, CA; Canada; Sweden; Japan; Belgium (2); Poland (2); Germany (4); Spain (4)
	NCT03611868 Angela Kaiser,Angela.Kaiser@asccentage.com,301-509-0357	A Study of APG-115 in Combination With Pembrolizumab in Patients With Metastatic Melanomas or Advanced Solid Tumors	Phase 1 /Phase 2	Saint Louis, MO; Philadelphia, PA; New York, NY; Hershey, PA; Santa Monica, CA; Durham, NC; Los Angeles, CA; Tucson, AZ; Houston, TX; Rogers, AR; Cleveland, OH; Fort Myers, FL; Washington, DC; San Antonio, TX; Nashville, TN; Fairfax, VA; Australia (4)
	NCT03878524 Kiara Siex, MPH,siex@ohsu.edu,503-418-3115	Serial Measurements of Molecular and Architectural Responses to Therapy (SMMART) PRIME Trial	Phase 1	Portland, OR
	NCT03964233 Boehringer Ingelheim,clintriage.rdg@boehringer-ingelheim.com,1-800-243-0127	A Study in Patients With Different Types of Advanced Cancer (Solid Tumors) to Test Different Doses of BI 907828 in Combination With Ezabenzimab With or Without BI 754111	Phase 1	Houston, TX; New York, NY; New Haven, CT; San Antonio, TX; Santa Monica, CA; Fairfax, VA; Netherlands; Singapore; Hungary; Belgium; Japan; United Kingdom; Australia; France (3); Germany (2); Spain (2)
	NCT04553133 Pfizer CT.gov Call Center,ClinicalTrials.gov_Inquiries@pfizer.com,1-800-718-1021	PF-07104091 as a Single Agent and in Combination Therapy	Phase 1 /Phase 2	Houston, TX; Grand Rapids, MI; Clive, IA; Middletown, NJ; Des Moines, IA (3); Boston, MA (3); Charlottesville, VA (3); NY (9); KY (11); Japan (2); China (2); Mexico (3)
	NCT04557449 Pfizer CT.gov Call Center,ClinicalTrials.gov_Inquiries@pfizer.com,1-800-718-1021	Study to Test the Safety and Tolerability of PF-07220060 in Participants With Advance Solid Tumors	Phase 1	Houston, TX; Grand Rapids, MI; Boston, MA (3); New Haven, CT (2); Nashville, TN (2); Argentina (3); Czechia (2); China (5); United Kingdom (4); Mexico (6); Slovakia (4)
ALK L1196Q	NCT02927340 Ibiayi Dagogo-Jack, MD,IDAGOGO-JACK@PARTNERS.ORG,617-724-4000	A Study of Lorlatinib in Advanced ALK and ROS1 Rearranged Lung Cancer With CNS Metastasis in the Absence of Measurable Extracranial Lesions	Phase 2	Boston, MA
	NCT03093116 Turning Point Therapeutics Medical Information,medinfo@tptherapeutics.com,+1 (855) 427-5878	A Study of Repotrectinib (TPX-0005) in Patients With Advanced Solid Tumors Harboring ALK, ROS1, or NTRK1-3 Rearrangements	Phase 1 /Phase 2	Appleton, WI; Saint Louis, MO; Baltimore, MD; Honolulu, HI; Memphis, TN; Yuma, AZ; Berkeley Heights, NJ; Houston, TX; Frederick, MD; Peoria, IL; Fairhaven, MA; Aurora, CO; Bolivar, MO; Saint Joseph, MO; Bellingham, WA; Belleville, NJ; Fairfax, VA; South Bend, IN; Goldsboro, NC; Philadelphia, PA; Bethesda, MD; Chicago, IL; New Brunswick, NJ; Columbus, GA; Jacksonville, FL; Tulsa, OK; Saint Paul, MN; Seattle, WA; Lansing, MI; Kingwood, TX; Fort Wayne, IN; Athens, GA; Hollywood, FL; Fredericksburg, VA; Cincinnati, OH; Gettysburg, PA; Dallas, TX; Ocala, FL; Salt Lake City, UT; Tampa, FL; Covington, LA; Detroit, MI (2); Boston, MA (2); Washington, DC (2);

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Alteration	Trial ID / Contact	Title	Phase	Site (number in parenthesis is count of trial sites)
				Sioux Falls, SD (2); New York, NY (2); Columbus, OH (2); CA (8); Denmark; Singapore (2); Hungary (2); Hong Kong (4); Japan (9); United Kingdom (4); Spain (8); Canada (5); Netherlands (2); Belgium (2); China (31); Taiwan (3); Poland (5); Korea, Republic of (10); Italy (7); France (8); Australia (3); Germany (4)
	NCT04227028 Victoria Villafior, vvillafior@coh.org, 626-359-8111	Brigatinib and Bevacizumab for the Treatment of ALK-Rearranged Locally Advanced, Metastatic, or Recurrent NSCLC	Phase 1	Duarte, CA; Madison, WI; Chicago, IL; Aurora, CO
	NCT04362072 Pfizer CT.gov Call Center, ClinicalTrials.gov_Inquiries@pfizer.com, 1-800-718-1021	Study of Lorlatinib In People With ALK-positive Non-small Cell Lung Cancer	Phase 4	Athens, GA; Orange, CA; Switzerland; Poland (7); United Kingdom (3); Italy (10); Spain (10); India (5)
	NCT04800822 Pfizer CT.gov Call Center, ClinicalTrials.gov_Inquiries@pfizer.com, 1-800-718-1021	PF-07284892 in Participants With Advanced Solid Tumors	Phase 1	Houston, TX; Detroit, MI; Rochester, MN; Novi, MI; Grand Rapids, MI; Scottsdale, AZ; Phoenix, AZ; Franklin, TN; New York, NY (2); Nashville, TN (2)
	NCT05097599 Stephanie Bush, stephanie.bush@strataoncology.com, (734) 527-1000	Strata PATH™ (Precision Indications for Approved Therapies)	Phase 2	Hot Springs, AR; Kettering, OH; Baton Rouge, LA; Milwaukee, WI; Madison, WI; West Palm Beach, FL; Fort Myers, FL; Iowa City, IA; Newark, DE; Columbus, OH; Saint Petersburg, FL; Tallahassee, FL; Chattanooga, TN
EML4-ALK Fusion	NCT02927340 Ibiayi Dagogo-Jack, MD, IDAGOGO-JACK@PARTNERS.ORG, 617-724-4000	A Study of Lorlatinib in Advanced ALK and ROS1 Rearranged Lung Cancer With CNS Metastasis in the Absence of Measurable Extracranial Lesions	Phase 2	Boston, MA
	NCT03093116 Turning Point Therapeutics Medical Information, medinfo@tptherapeutics.com, +1 (855) 427-5878	A Study of Repotrectinib (TPX-0005) in Patients With Advanced Solid Tumors Harboring ALK, ROS1, or NTRK1-3 Rearrangements	Phase 1 /Phase 2	Appleton, WI; Saint Louis, MO; Baltimore, MD; Honolulu, HI; Memphis, TN; Yuma, AZ; Berkeley Heights, NJ; Houston, TX; Frederick, MD; Peoria, IL; Fairhaven, MA; Aurora, CO; Bolivar, MO; Saint Joseph, MO; Bellingham, WA; Belleville, NJ; Fairfax, VA; South Bend, IN; Goldsboro, NC; Philadelphia, PA; Bethesda, MD; Chicago, IL; New Brunswick, NJ; Columbus, GA; Jacksonville, FL; Tulsa, OK; Saint Paul, MN; Seattle, WA; Lansing, MI; Kingwood, TX; Fort Wayne, IN; Athens, GA; Hollywood, FL; Fredericksburg, VA; Cincinnati, OH; Gettysburg, PA; Dallas, TX; Ocala, FL; Salt Lake City, UT; Tampa, FL; Covington, LA; Detroit, MI (2); Boston, MA (2); Washington, DC (2); Sioux Falls, SD (2); New York, NY (2); Columbus, OH (2); CA (8); Denmark; Singapore (2); Hungary (2); Hong Kong (4); Japan (9); United Kingdom (4); Spain (8); Canada (5); Netherlands (2); Belgium (2); China (31); Taiwan (3); Poland (5); Korea, Republic of (10); Italy (7); France (8); Australia (3); Germany (4)
	NCT04227028	Brigatinib and Bevacizumab for the Treatment	Phase 1	Duarte, CA; Madison, WI; Chicago, IL;

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List of Available Clinical Trials

Alteration	Trial ID / Contact	Title	Phase	Site (number in parenthesis is count of trial sites)
	Victoria Villafior,vvillafior@coh.org,626-359-8111	of ALK-Rearranged Locally Advanced, Metastatic, or Recurrent NSCLC		Aurora, CO
	NCT04362072 Pfizer CT.gov Call Center,ClinicalTrials.gov_Inquiries@pfizer.com,1-800-718-1021	Study of Lorlatinib In People With ALK-positive Non-small Cell Lung Cancer	Phase 4	Athens, GA; Orange, CA; Switzerland; Poland (7); United Kingdom (3); Italy (10); Spain (10); India (5)
	NCT04800822 Pfizer CT.gov Call Center,ClinicalTrials.gov_Inquiries@pfizer.com,1-800-718-1021	PF-07284892 in Participants With Advanced Solid Tumors	Phase 1	Houston, TX; Detroit, MI; Rochester, MN; Novi, MI; Grand Rapids, MI; Scottsdale, AZ; Phoenix, AZ; Franklin, TN; New York, NY (2); Nashville, TN (2)
	NCT05097599 Stephanie Bush,stephanie.bush@strataoncology.com,(734) 527-1000	Strata PATH™ (Precision Indications for Approved Therapies)	Phase 2	Hot Springs, AR; Kettering, OH; Baton Rouge, LA; Milwaukee, WI; Madison, WI; West Palm Beach, FL; Fort Myers, FL; Iowa City, IA; Newark, DE; Columbus, OH; Saint Petersburg, FL; Tallahassee, FL; Chattanooga, TN

Detailed Therapy Results

Alteration	Drug	Trade Name	Target	Current Status
CDKN2A R80*	Abemaciclib	Verzenio	Cdk4/6/9 inhibitor.	Phase 3 (Non-small cell lung carcinoma (NSCLC)) FDA Approved in other indications (Breast carcinoma (hormone receptor +, HER2-))
	ALRN-6924		Dual Mdm2/Mdm4 inhibitor, reactivates wild-type p53.	Phase 1 (Solid Tumor) Phase 1 (Cancer)
	Alvocidib		Cdk inhibitor targeting several Cdk, including Cdk1, Cdk2, Cdk4, Cdk5, Cdk6, Cdk7, and Cdk9.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 2 (Gastric carcinoma, Hepatocellular carcinoma (HCC), Pancreatic carcinoma, Melanoma, Prostate carcinoma, Endometrial carcinoma, Renal cell carcinoma, Ovarian carcinoma, Head and neck carcinoma, Gastroesophageal junction carcinoma, Germ cell tumor, Esophageal carcinoma, Breast carcinoma, Chronic lymphocytic leukemia (CLL), Acute myeloid leukemia (AML), Multiple myeloma (MM), B-cell lymphoma, Testicular cancer, Lung cancer, Sarcoma, Colorectal carcinoma (CRC), Acute lymphoblastic leukemia (ALL), Myelodysplastic Syndrome (MDS))
	AMG 232		Small molecule inhibitor of the Mdm2-p53 interaction.	Phase 1 (Solid Tumor) Phase 2 (Melanoma)
	APG-115		Small molecule inhibitor of the Mdm2-p53 interaction.	Phase 1 (Solid Tumor) Phase 2 (Melanoma)
	BI 907828		Small molecule inhibitor of the Mdm2-p53 interaction.	Phase 1 (Solid Tumor)
	Dalpiciclib		Cdk4/6 inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 3 (Breast carcinoma)
	FN-1501		Multikinase inhibitor of Cdk2/4/6 and Flt3.	Phase 1 (Solid Tumor) Phase 1 (Acute myeloid leukemia (AML))
	HDM201		Inhibitor of the Mdm2-p53 interaction.	Phase 1 (Solid Tumor) Phase 2 (Liposarcoma)
	Kevetrin		Blocks Mdm2-p53 interaction, restoring transcriptional activity of p53.	Phase 1 (Solid Tumor) Phase 2 (Ovarian carcinoma)
	Lerociclib		Cdk4/6 inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 2 (Breast carcinoma)
	MK-8242		Small molecule inhibitor of the Mdm2-p53 interaction.	Phase 1 (Solid Tumor) Phase 1 (Acute myeloid leukemia (AML), Liposarcoma)
	Narazaciclib		Ark5 and Cdk4 multi-kinase inhibitor.	Phase 1 (Solid Tumor)
	Palbociclib	Ibrance	Cdk4/6 inhibitor.	Phase 3 (Non-small cell lung carcinoma (NSCLC)) FDA Approved in other indications (Breast carcinoma (hormone receptor +, HER2-))
	PF-06873600		Cdk2/4/6 inhibitor.	Phase 2 (Ovarian carcinoma, Breast carcinoma)
	PF-07220060		Cdk4 inhibitor.	Phase 1 (Non-small cell lung carcinoma (NSCLC)) Phase 1 (Solid Tumor, Liposarcoma)
	Ribociclib	Kisqali	Cdk4/6 inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) FDA Approved in other indications (Breast carcinoma (hormone receptor +, HER2-))

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Detailed Therapy Results

Alteration	Drug	Trade Name	Target	Current Status
	STEMVAC		CD105 (ENG)/Yb-1/SOX2/CDH3 /MDM2 vaccine.	Phase 1 (Breast carcinoma)
	TQB3616		Cdk4/6 inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 2 (Breast carcinoma)
	Trilaciclib	Cosela	Cdk4/6 inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) FDA Approved in other indications (Small cell lung carcinoma (SCLC))
	Voruciclib		Multikinase inhibitor of Cdk1/4/6/9.	Phase 1 (Solid Tumor) Phase 1 (Marginal zone lymphoma (MZL), Brain and Central Nervous System Tumors, Hematologic malignancies, Chronic lymphocytic leukemia (CLL), Acute myeloid leukemia (AML), Mantle cell lymphoma (MCL), Follicular lymphoma (FL), Diffuse large B-cell lymphoma (DLBCL), Small lymphocytic lymphoma)
ALK L1196Q	Alkotinib		Dual Alk/Ros1 inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC))
	Brigatinib	Alunbrig	Dual Alk/Egfr kinase inhibitor.	Phase 3 (Non-small cell lung carcinoma (NSCLC)) FDA Approved in other indications (ALK-rearranged NSCLC)
	Ceritinib	Zykadia	Alk/Ros1 kinase inhibitor.	Phase 3 (Non-small cell lung carcinoma (NSCLC)) FDA Approved in other indications (ALK-rearranged NSCLC)
	Ensartinib		Alk small molecule kinase inhibitor.	Phase 3 (Non-small cell lung carcinoma (NSCLC)) Phase 3 (Lung cancer)
	Entrectinib	Rozlytrek	Pan-Trk, Ros1, and Alk kinase inhibitor.	Phase 3 (Non-small cell lung carcinoma (NSCLC)) FDA Approved in other indications (Solid tumor with NTRK fusion, ROS1-rearranged NSCLC)
	Lorlatinib	Lorbrena	Alk/Ros1 inhibitor.	Phase 3 (Non-small cell lung carcinoma (NSCLC)) FDA Approved in other indications (ALK-rearranged NSCLC)
	PLB1003		Alk inhibitor.	Phase 1 (Non-small cell lung carcinoma (NSCLC))
	Repotrectinib		Alk/Ros1/Src/Trk/FAK kinase inhibitor.	Phase 2 (Solid Tumor) Phase 2 (Brain and Central Nervous System Tumors, Non-Hodgkin lymphoma (NHL))
	SY-3505		Alk inhibitor.	Phase 1 (Non-small cell lung carcinoma (NSCLC))
	TPX-0131		Alk inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC))
	WX-0593		Alk inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 1 (Solid Tumor, Brain and Central Nervous System Tumors)
EML4-ALK Fusion	Alkotinib		Dual Alk/Ros1 inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC))
	Brigatinib	Alunbrig	Dual Alk/Egfr kinase inhibitor.	Phase 3 (Non-small cell lung carcinoma (NSCLC)) FDA Approved in other indications (ALK-rearranged NSCLC)
	Ceritinib	Zykadia	Alk/Ros1 kinase inhibitor.	Phase 3 (Non-small cell lung carcinoma (NSCLC)) FDA Approved in other indications (ALK-rearranged NSCLC)
	Ensartinib		Alk small molecule kinase inhibitor.	Phase 3 (Non-small cell lung carcinoma (NSCLC)) Phase 3 (Lung cancer)
	Entrectinib	Rozlytrek	Pan-Trk, Ros1, and Alk kinase	Phase 3 (Non-small cell lung carcinoma (NSCLC))

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Detailed Therapy Results

Alteration	Drug	Trade Name	Target	Current Status
			inhibitor.	FDA Approved in other indications (Solid tumor with NTRK fusion, ROS1-rearranged NSCLC)
	Lorlatinib	Lorbrena	Alk/Ros1 inhibitor.	Phase 3 (Non-small cell lung carcinoma (NSCLC)) FDA Approved in other indications (ALK-rearranged NSCLC)
	PLB1003		Alk inhibitor.	Phase 1 (Non-small cell lung carcinoma (NSCLC))
	Repotrectinib		Alk/Ros1/Src/Trk/FAK kinase inhibitor.	Phase 2 (Solid Tumor) Phase 2 (Brain and Central Nervous System Tumors, Non-Hodgkin lymphoma (NHL))
	SY-3505		Alk inhibitor.	Phase 1 (Non-small cell lung carcinoma (NSCLC))
	TPX-0131		Alk inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC))
	WX-0593		Alk inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 1 (Solid Tumor, Brain and Central Nervous System Tumors)

Relevance of Detected Alterations

Alteration	Role in Disease	Effect on Drug Sensitivity	Effect on Drug Resistance
ALK L1196Q	ALK was originally identified in anaplastic lymphoma as a fusion partner with the gene product of NPM1; ALK has subsequently been identified as a fusion partner with numerous other genes, including EML4 in lung cancer. ⁽¹⁾ . The ALK gene can become oncogenic by a gene rearrangement, copy number gain, or genetic mutation. ^(1,2) . Patients with EML4-ALK fusions generally have wild-type EGFR, KRAS, and TP53, and are resistant to Egfr inhibitors, although there have been reports of non-small cell lung carcinoma (NSCLC) tumors which harbor concomitant EGFR mutations and EML4-ALK translocations. ^(3,4) . ALK rearrangements have been associated with younger age, increased disease stage, nodal metastasis, invasive mucinous disease, and with epithelial-mesenchymal transition marker expression in lung adenocarcinoma. ⁽⁵⁻¹⁰⁾ .	Tumors with ALK activation, by either mutation, fusion, or amplification, may be sensitive to Alk inhibitors. ⁽¹¹⁾ . The Alk inhibitors crizotinib, ceritinib, alectinib, brigatinib, and lorlatinib have been approved for ALK-translocation-positive non-small cell lung cancer; additional Alk inhibitors are under investigation in clinical trials. ⁽¹²⁻¹⁷⁾ . Crizotinib has additionally been FDA-approved for the treatment of pediatric and young adult patients with relapsed /refractory ALK-positive anaplastic large cell lymphoma or ALK-positive inflammatory myofibroblastic tumors. ⁽¹⁸⁻²⁰⁾ . However, case studies have reported emergence of ALK L1196Q as a secondary mutation in ALK-rearranged patients following resistance to alectinib and crizotinib. ⁽²¹⁻²³⁾ .	Secondary mutations in ALK can confer resistance to different Alk inhibitors, such as crizotinib, ceritinib, alectinib, or brigatinib, in ALK-rearranged NSCLC. ⁽²⁴⁻²⁸⁾ . Other mechanisms of resistance to crizotinib have also been reported, including activation of Egfr, K-Ras, PI3K/Akt /mTOR, and Hsp90 signaling. ⁽²⁹⁻³⁶⁾ . Multiple case studies have reported that failure of alectinib treatment in NSCLC harboring ALK rearrangements is associated with transformation of disease into SCLC. ⁽³⁷⁻³⁹⁾ .
EML4-ALK Fusion	ALK was originally identified in anaplastic lymphoma as a fusion partner with the gene product of NPM1; ALK has subsequently been identified as a fusion partner with numerous other genes, including EML4 in lung cancer. ⁽¹⁾ . The ALK gene can become oncogenic by a gene rearrangement, copy number gain, or genetic mutation. ^(1,2) . Patients with EML4-ALK fusions generally have wild-type EGFR, KRAS, and TP53, and are resistant to Egfr inhibitors, although there have been reports of non-small cell lung carcinoma (NSCLC) tumors which harbor concomitant EGFR mutations and EML4-ALK translocations. ^(3,4) . ALK rearrangements have been associated with younger age, increased disease stage, nodal metastasis, invasive mucinous disease, and with epithelial-mesenchymal transition marker expression in lung adenocarcinoma. ⁽⁵⁻¹⁰⁾ .	Tumors with ALK activation, by either mutation, fusion, or amplification, may be sensitive to Alk inhibitors. ⁽¹¹⁾ . The Alk inhibitors crizotinib, ceritinib, alectinib, brigatinib, and lorlatinib have been approved for ALK-translocation-positive non-small cell lung cancer; additional Alk inhibitors are under investigation in clinical trials. ⁽¹²⁻¹⁷⁾ . Crizotinib has additionally been FDA-approved for the treatment of pediatric and young adult patients with relapsed /refractory ALK-positive anaplastic large cell lymphoma or ALK-positive inflammatory myofibroblastic tumors. ⁽¹⁸⁻²⁰⁾ . In this case, the presence of a crizotinib- and alectinib-resistance mutation suggests that this tumor may be resistant to these drugs.	Secondary mutations in ALK can confer resistance to different Alk inhibitors, such as crizotinib, ceritinib, alectinib, or brigatinib, in ALK-rearranged NSCLC. ⁽²⁴⁻²⁸⁾ . Other mechanisms of resistance to crizotinib have also been reported, including activation of Egfr, K-Ras, PI3K/Akt /mTOR, and Hsp90 signaling. ⁽²⁹⁻³⁶⁾ . Multiple case studies have reported that failure of alectinib treatment in NSCLC harboring ALK rearrangements is associated with transformation of disease into SCLC. ⁽³⁷⁻³⁹⁾ .
CDKN2A R80*	Because the CDKN2A and CDKN2B gene products encode proteins that act as tumor suppressors, deletion or loss of activity may result in deregulation of the p16INK4a/Cdk4 /Cyclin/Rb and/or the Mdm2/p53 pathways, and altered regulation of the	There are currently no drugs that directly target inactivating mutations or loss of CDKN2A. Because p16INK4a is known to inhibit Cdk4, tumors with CDKN2A alterations may be sensitive to Cdk4/6 inhibitors. ^(41,51) . p14ARF	Inactivating CDKN2A alterations, most commonly deletion, have been significantly associated with resistance or lack of response to immune checkpoint blockade monotherapy in NSCLC patients; however, this association was not observed in

Relevance of Detected Alterations

Alteration	Role in Disease	Effect on Drug Sensitivity	Effect on Drug Resistance
	cell cycle. ^(40,41) . Loss of p16INK4a expression is thought to be an early event in NSCLC. ^(42,43) . One study reported that loss of CDKN2A and CDKN2B in the context of KRAS mutation accelerated the formation and progression of lung adenocarcinoma tumors in mice, as compared with KRAS mutation alone, resulting in high-grade tumors with lymph node metastasis. ⁽⁴⁴⁾ . Inactivating CDKN2A alterations or low mRNA expression has been significantly associated with brain metastasis in NSCLC patients. ⁽⁴⁵⁻⁴⁸⁾ . In addition, CDKN2A inactivation or low expression has been significantly associated with low expression of immune checkpoint inhibitor-related genes, including CD274/PD-L1, in NSCLC cases. ^(49,50) .	has been reported to function as a tumor suppressor through stabilization and activation of p53, via a mechanism of Mdm2 inhibition. ^(41,53) . Trials of Mdm2 antagonists in solid tumors and hematological malignancies have shown clinical efficacy, but also some on-target toxicity. ⁽⁵⁴⁾ .	patients treated with immune checkpoint blockade plus chemotherapy. ^(48,50) .

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[REDACTED] (A0726772)

[REDACTED] Test Number 1

(A0726772)

Patient MRN: 007087315 | Sex: Male

Diagnosis: Lung adenocarcinoma

REPORTING

Report Date: APR-19-2023
Receipt Date: APR-14-2023
Collection Date: APR-13-2023
Specimen: Blood
Status: FINAL

PHYSICIAN

Susanne Arnold
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Additional Recipient: N/A

Companion Diagnostic

No reportable alterations with companion diagnostic (CDx) claims

Other Biomarkers Identified

Results reported in this section are not prescriptive or conclusive for labeled use of any specific therapeutic product. See professional services section for additional information.

Biomarkers with Evidence of Clinical Significance in Tissue and ctDNA [†]

Biomarker	Status	Additional Information
ALK fusions	DETECTED	EML4-ALK Fusion See professional services section for additional information

[†]Please refer below to Performance Characteristics and Definitions sections for descriptions of categories.

Other Biomarkers with Potential Clinical Significance

Clinical significance has not yet been established for biomarkers in this section. See the professional services section for additional information.

– ALK L1196Q – CDKN2A R80*

Intended Use

Guardant360® CDx is a qualitative next generation sequencing-based in vitro diagnostic device that uses targeted high throughput hybridization-based capture technology for detection of single nucleotide variants (SNVs), insertions and deletions (indels) in 55 genes, copy number amplifications (CNAs) in two (2) genes, and fusions in four (4) genes. Guardant360 CDx utilizes circulating cell-free DNA (cfDNA) from plasma of peripheral whole blood collected in Streck Cell-Free DNA Blood Collection Tubes (BCTs). The test is intended to be used as a companion diagnostic to identify patients who may benefit from treatment with the therapies listed in **Table 1** in accordance with the approved therapeutic product labeling.

Table 1. Companion Diagnostic Indications

Indication	Biomarker	Therapy
Non-small cell lung cancer (NSCLC)	<i>EGFR</i> exon 19 deletions, L858R, and T790M*	TAGRISSO® (osimertinib)
	<i>EGFR</i> exon 20 insertions	RYBREVENT® (amivantamab-vmjw)
	<i>ERBB2/HER2</i> activating mutations (SNVs and exon 20 insertions)	ENHERTU® (fam-trastuzumab deruxtecan-nxki)
	<i>KRAS</i> G12C	LUMAKRAS™ (sotorasib)
Breast cancer	<i>ESR1</i> missense mutations between codons 310 and 547	ORSERDU™ (elacestrant)

A negative result from a plasma specimen does not assure that the patient's tumor is negative for genomic findings. Patients who are negative for the biomarkers listed in **Table 1** should be reflexed to tissue biopsy testing for **Table 1** biomarkers using an FDA-approved tumor tissue test, if feasible.

*The efficacy of TAGRISSO (osimertinib) has not been established in the *EGFR* T790M plasma-positive, tissue-negative or unknown population and clinical data for T790M plasma-positive patients are limited; therefore, testing using plasma specimens is most appropriate for consideration in patients from whom a tumor biopsy cannot be obtained.

Additionally, the test is intended to provide tumor mutation profiling to be used by qualified health care professionals in accordance with professional guidelines in oncology for cancer patients with any solid malignant neoplasm. The test is for use with patients previously diagnosed with cancer and in conjunction with other laboratory and clinical findings.

Genomic findings other than those listed in **Table 1** are not prescriptive or conclusive for labeled use of any specific therapeutic product.

Guardant360 CDx is a single-site assay performed at Guardant Health, Inc.

Warnings and Precautions

- Alterations reported may include somatic (not inherited) or germline (inherited) alterations. The assay filters germline variants from reporting except for pathogenic *BRCA1*, *BRCA2*, *ATM*, and *CDK12* alterations. However, if a reported alteration is suspected to be germline, confirmatory testing should be considered in the appropriate clinical context.
- The test is not intended to replace germline testing or to provide information about cancer predisposition.
- Somatic alterations in *ATM* and *CDK12* are not reported by the test as they are excluded from the test's reportable range.
- Genomic findings from cfDNA may originate from circulating tumor DNA (ctDNA) fragments, germline alterations, or non-tumor somatic alterations, such as clonal hematopoiesis of indeterminate potential (CHIP).
- Allow the tube to fill completely until blood stops flowing into the tube. Underfilling of tubes with less than 5 mL of blood (bottom of the label indicates 5 mL fill when tube is held vertically) may lead to incorrect analytical results or poor product performance. This tube has been designed to fill with 10 mL of blood.

Limitations

- For *in vitro* diagnostic use.
- For prescription use only. This test must be ordered by a qualified medical professional in accordance with clinical laboratory regulations.
- The efficacy of TAGRISSO (osimertinib) has not been established in the *EGFR* T790M plasma-positive, tissue-negative or unknown population, and clinical data for T790M plasma-positive patients are limited; therefore, testing using plasma specimens is most appropriate for consideration in patients from whom a tumor biopsy cannot be obtained.
- TAGRISSO efficacy has not been established in patients with *EGFR* exon 19 deletions < 0.08% MAF, in patients with *EGFR* L858R < 0.09% MAF, and in patients with *EGFR* T790M < 0.03% MAF.
- RYBREVENT efficacy has not been established in patients with *EGFR* exon 20 insertions < 0.02% MAF.
- LUMAKRAS efficacy has not been established in patients with *KRAS* G12C biomarkers < 0.11% MAF.
- ENHERTU efficacy has not been established in patients with *ERBB2* exon 20 insertions < 0.03% MAF and in patients with *ERBB2* SNVs < 0.23% MAF.
- ORSERDU efficacy has not been established in patients with *ESR1* missense mutations < 0.03% MAF.
- The test is not intended to be used for standalone diagnostic purposes.
- The test is intended to be performed on specific serial number-controlled instruments by Guardant Health, Inc.
- A negative result for any given variant does not preclude the presence of this variant in tumor tissue.
- Decisions on patient care and treatment must be based on the independent medical judgment of the treating physician, taking into consideration all applicable information concerning the patient's condition, such as patient and family history, physical examinations, information from other diagnostic tests, and patient preferences, in accordance with the standard of care.
- ctDNA shedding rate may be lower in patients with primary central nervous system (CNS) tumors.

Performance Characteristics

Please refer to product label, www.guardant360cdx.com/technicalinfo. Clinical Performance has not been established for biomarkers in categories 2, 3A, 3B and 4. Guardant360 CDx is indicated to report the following SNVs (*AKT1*, *ALK*, *APC*, *AR*, *ARAF*, *ATM*[#], *BRAF*, *BRCA1*^{##}, *BRCA2*^{##}, *CCND1*, *CDH1*, *CDK4*, *CDK6*, *CDK12*[#], *CDKN2A*, *CTNNB1*, *EGFR*, *ERBB2*, *ESR1*, *FGFR1*, *FGFR2*, *FGFR3*, *GATA3*, *GNA11*, *GNAQ*, *HRAS*, *IDH1*, *IDH2*, *KIT*, *KRAS*, *MAP2K1*, *MAP2K2*, *MET*, *MLH1*, *MTOR*, *MYC*, *NF1*, *NFE2L2*, *NRAS*, *NTRK1*, *NTRK3*, *PDGFRA*, *PIK3CA*, *PTEN*, *RAF1*, *RET*, *RHEB*, *ROS1*, *SMAD4*, *SMO*, *STK11*, *TERT*, *TSC1*, *VHL*), Indels (*ALK*, *AKT1*, *APC*, *ATM*[#], *BRAF*, *BRCA1*^{##}, *BRCA2*^{##}, *CDH1*, *CDK12*[#], *CDKN2A*, *EGFR*, *ERBB2*, *ESR1*, *FGFR2*, *GATA3*, *HNFI1A*, *HRAS*, *KIT*, *KRAS*, *MET*, *MLH1*, *NF1*, *PDGFRA*, *PIK3CA*, *PTEN*, *RET*, *ROS1*, *STK11*, *TSC1*, *VHL*), Fusion (*ALK*, *NTRK1*, *RET*, *ROS1*), and Amplifications (*ERBB2*, *MET*).

[#]Reporting is enabled for pathogenic germline alterations only. Somatic alterations will not be reported. | ^{##}Reporting is enabled for both germline and somatic alterations.

(A0726772)

Patient MRN: 007087315 | Sex: Male

Diagnosis: Lung adenocarcinoma

Definition of Categories

The test report includes genomic finding reported in the following categories:

Category	Prescriptive use for Therapeutic Product	Clinical Performance	Analytical Performance	Comments
Category 1: Companion Diagnostic (CDx)	Yes	Yes	Yes	ctDNA biomarkers linked to the safe and effective use of the corresponding therapeutic product, for which Guardant360 CDx has demonstrated clinical performance shown to support therapeutic efficacy and strong analytical performance for the biomarker.
Category 2: ctDNA Biomarkers with Strong Evidence of Clinical Significance in ctDNA	No	No	Yes	ctDNA biomarkers with strong evidence of clinical significance presented by other FDA-approved liquid biopsy companion diagnostics for which Guardant360 CDx has demonstrated analytical reliability but not clinical performance.
Category 3A: Biomarkers with Evidence of Clinical Significance in tissue supported by: strong analytical validation using ctDNA	No	No	Yes	ctDNA biomarkers with evidence of clinical significance presented by tissue-based FDA-approved companion diagnostics or professional guidelines for which Guardant360 CDx has demonstrated analytical performance including analytical accuracy, and concordance of blood-based testing to tissue-based testing for the biomarker.
Category 3B: Biomarkers with Evidence of Clinical Significance in tissue supported by: analytical validation using ctDNA	No	No	Yes	ctDNA biomarkers with evidence of clinical significance presented by tissue-based FDA-approved companion diagnostics or professional guidelines for which Guardant360 CDx has demonstrated minimum analytical performance including analytical accuracy.
Category 4: Other Biomarkers with Potential Clinical Significance	No	No	Yes	ctDNA biomarkers with emergent evidence based on peer-reviewed publications for genes/variants in tissue, variant information from well- curated public databases, or <i>in-vitro</i> pre-clinical models, for which Guardant360 CDx has demonstrated minimum analytical performance.

Testing performed at: Guardant Health

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